ARISTA MAR-ACC-30 MARATHAHALLI CORE ACCESS NODE OPTICAL TRANSPORT FAILURE STANDARD OPERATING PROCEDURE

Node: MAR-ACC-30 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_2320

Description

This Standard Operating Procedure (SOP) is designed to address the occurrence of an Optical Transport Failure on the MAR-ACC-30 node located at the Marathahalli Core site, which is an Access node utilizing Arista vendor equipment. The failure domain in question pertains to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss. The ALARM_2320 alarm code with a severity of Warning is generated in response to these conditions. This failure domain arises due to the intricacies of the control and data plane protocols and hardware mechanics employed in the Arista equipment. Specifically, the Ethernet Operations, Administration, and Maintenance (EOAM) protocol and the Open System Interconnection (OSI) model's Data Link Layer (Layer 2) are involved in the Optical Transport Failure. The EOAM protocol facilitates the monitoring of Ethernet network performance and is responsible for generating alarm notifications when issues arise. The Optical Transport Failure is also influenced by the hardware mechanics of the SFP/XFP transceivers, which convert optical signals to electrical signals and vice versa. The degradation of these transceivers or fiber cuts can lead to a loss of signal strength, resulting in Rx/Tx power loss. As a result, the Arista equipment generates the ALARM_2320 alarm code, indicating a potential issue that requires immediate attention.

Pre-requisite Checks

- Verify that the MAR-ACC-30 node is located at the Marathahalli Core site and that the Access node is functioning as expected.

- Check the redundant link status to ensure that it is operational and not contributing to the Optical Transport Failure.

- Dispatch remote hands to the Marathahalli Core site to assist in the diagnosis and resolution of the issue.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cut	Medium
Rx/Tx power loss due to hardware or software issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces status, Expected Output: Interface Status: 100GE1/0/0, Admin State: up, Operational State: down, Description: This command is used to verify the operational status of the interface and identify any potential issues that may be contributing to the Optical Transport Failure.

Command: show system internal alarm, Expected Output: ALARM_2320: Optical Transport Failure, Description: This command is used to verify the presence of the ALARM_2320 alarm code and confirm that it is related to the Optical Transport Failure.

Command: show interface statistics, Expected Output: Input Packets: 0, Input Errors: 0, Output Packets: 0, Output Errors: 0, Description: This command is used to verify the packet count and error statistics for the interface and identify any potential issues that may be contributing to the Optical Transport Failure.
```

Resolution Steps

Step 1: Step: Verify SFP/XFP transceiver status, Command: show interfaces transceiver, Description: This command is used to verify the operational status of the SFP/XFP transceiver and check for any degradation or issues that may be contributing to the Optical Transport Failure.

Step 2: Step: Replace SFP/XFP transceiver, Command: hardware replace transceiver SFP-100G-SR4, Description: This command is used to replace the degraded SFP/XFP transceiver with a new one and restore the interface to an operational state.

Step 3: Step: Verify fiber connectivity, Command: show interfaces fiber, Description: This command is used to verify the operational status of the fiber connections and check for any issues that may be contributing to the Optical Transport Failure.

Post-Check Verification

```
Command: show interfaces status, Expected Output: Interface Status: 100GE1/0/0, Admin State: up, Operational State: up, Description: This command is used to verify that the interface is now operational and that the Optical Transport Failure has been resolved.

Command: show system internal alarm, Expected Output: , Description: This command is used to verify that the ALARM_2320 alarm code is no longer present and that the Optical Transport Failure has been resolved.
```

Roll-back Procedures

- Step: Revert SFP/XFP transceiver replacement, Command: hardware revert transceiver SFP-100G-SR4, Description: This command is used to revert the SFP/XFP transceiver replacement and restore the original transceiver in case the new one does not resolve the issue.

- Step: Re-verify fiber connectivity, Command: show interfaces fiber, Description: This command is used to re-verify the operational status of the fiber connections and check for any issues that may be contributing to the Optical Transport Failure.